

* ϕ^3 at order ϕ^3 :

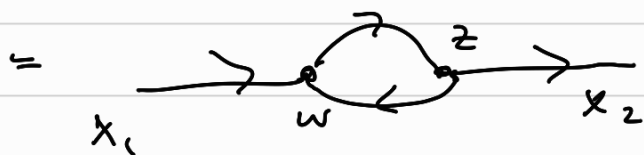
$$\langle \Omega | T \{ \phi_1 \phi_2 \} | \Omega \rangle = \frac{\langle 0 | \phi_1 \phi_2 e^{i \int \mathcal{L}_{int}} | 0 \rangle}{\langle 0 | e^{i \int \mathcal{L}_{int}} | 0 \rangle}$$

$$= \langle \Omega | T \phi_1 \phi_2 | \Omega \rangle_c + \text{Disjoint}$$

$$\langle 0 | \phi_1 \phi_2 (\mathcal{L} + i g \int d^4 w - g^2 \int d^4 w d^4 z) | 0 \rangle$$

$$g^2 \sim g^2 \int d^4 w d^4 z \langle 0 | \phi_1 \phi_2 \phi_w^3 \phi_z^3 | 0 \rangle$$

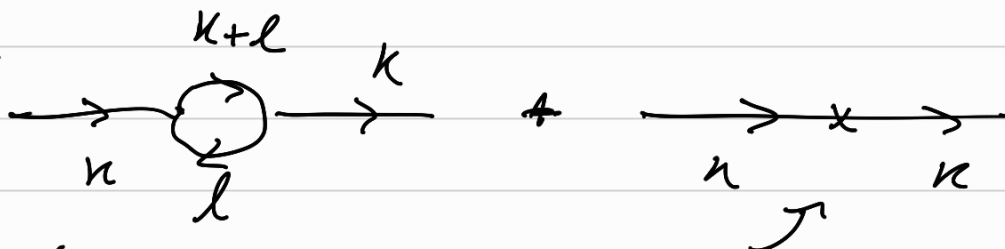
$$\rightarrow \langle 0 | \phi_1 \phi_w \phi_2 \phi_z \phi_w \phi_w \phi_z \phi_z | 0 \rangle$$



$$+ x_1 \longrightarrow x_2 \left(\bigcirc + \bigcirc - \bigcirc \right)$$

$$\langle \Omega | T \{ \phi_1 \phi_2 \} | \Omega \rangle = \text{diagram with a loop} + \dots$$

* Thus, at $\mathcal{O}(g^2)$ we have the following



$$\int \frac{d^d l}{(2\pi)^3} \frac{1}{(-ig) (k+l)^2 + m^2 - i\epsilon} (ig) \frac{1}{l^2 + m^2 - i\epsilon}$$

proof: $\langle 0 | \phi_1 \phi_w \phi_2 \phi_w \phi_z \phi_z \phi_z \phi_z | 0 \rangle$

$$= \int d^4 w d^4 z \Delta_{1w} \Delta_{zw}^2 \Delta_{zz}$$

$$= \int d^4 z d^4 w \left(\int \frac{d^4 k}{(2\pi)^4} \frac{e^{ik_1(x_1 - x_2)}}{k^2 + m^2 - i\epsilon} \right) \left(\int \frac{d^4 p}{(2\pi)^4} \frac{e^{ip_1 w}}{p^2 + m^2 - i\epsilon} \right)$$

$$\text{w.r.t. } \int d^4 z \left(\int d^4 w e^{-ik_1 w} e^{-ip_1 w} e^{ip_2 w} \right)$$

$$= \int d^4 z (2\pi)^d \delta^d(P_2 - k_i - p_1)$$

$$\underline{P_2 = k_i + l}, \quad \underline{l \equiv p_1}$$

$$= \int \frac{d^4 k_i}{(2\pi)^d} \frac{d^4 k_f}{(2\pi)^d} \frac{d^4 l}{(2\pi)^d} e^{ik_i x_1} e^{-ik_f x_2} \tilde{\Delta}(k_i) \tilde{\Delta}(l)$$

$$\times \int d^4 z e^{ik_f z} e^{ip_1 z} e^{-ip_2 z} \tilde{\Delta}(k_f) \tilde{\Delta}(k_i + l)$$

$$= (2\pi)^d \delta^d(k_f + p_1 - p_2)$$

$$k_f = p_2 - p_1 = k_i + l - l = k_i$$

$$\boxed{k_f = k_i \equiv k}$$

$$\int \frac{d^4 k}{(2\pi)^d} \int \frac{d^4 l}{(2\pi)^d} e^{ik(x_1 - x_2)} \tilde{\Delta}(k^2) \tilde{\Delta}(l^2) \tilde{\Delta}(k+l)^2$$

* Thus, we have shown that the $\mathcal{O}(g^2)$ loop correction to ϕ^3 theory is given by

$$\frac{1}{i} \tilde{\Delta}(k^2) = \frac{1}{i} \tilde{\Delta}(k^2) + \frac{1}{i} \tilde{\Delta}(k^2) [i\pi(k^2)] \tilde{\Delta}(k^2)$$

$$\tilde{\Delta}(k^2) \equiv \frac{1}{k^2 + m^2 - i\epsilon}$$

and

$$\Sigma(i\pi(k^2)) = \frac{1}{2} (ig)^2 \left(\frac{1}{i}\right)^2 \int \frac{d^d l}{(2\pi)^d} \tilde{\Delta}((l+k)^2) \tilde{\Delta}(l^2) - i(Ak^2 + Bm^2) + \mathcal{O}(g^4)$$

(Self-energy)

* The $-i(Ak^2 + Bm^2)$ are counter vertex terms that connect two prop. each w/ the same 4-momenta

$$A = Z_p - 1$$

$$B = Z_m - 1$$

* At higher order we have

$$\text{---} + \text{---} (\pi) \text{---} + \text{---} (\pi) \text{---} (\pi) \text{---}$$

$$\frac{1}{i} \hat{\Delta}(k^2) = \frac{1}{i} \tilde{\Delta}(k^2) + \frac{1}{i} \tilde{\Delta}(k^2) (i\pi(k^2)) \frac{1}{i} \tilde{\Delta}(k^2) + \frac{1}{i} \tilde{\Delta}(k^2) [i\pi(k^2)] \frac{1}{i} \tilde{\Delta}(k^2) [i\pi(k^2)] \frac{1}{i} \tilde{\Delta}(k^2) + \dots$$

$$\frac{1}{x} \left(\frac{x}{1 + \frac{x}{a}} \right) = \frac{1}{a} = \frac{x}{a^2} + \frac{x^2}{a^3} + \dots$$

Here $\frac{1}{a} = \tilde{\Delta}(k^2)$ and $x = -(i\pi)$

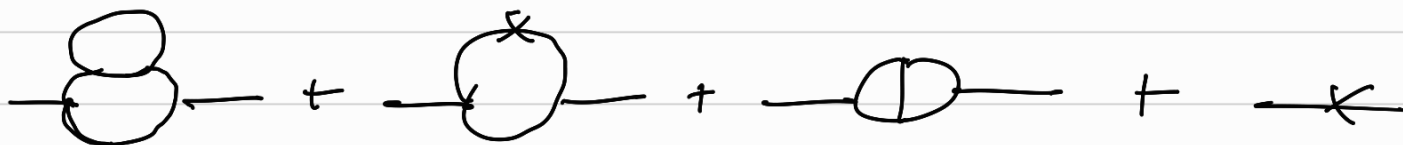
$$\frac{1}{k^2 + (k^2 - i\pi) - i\varepsilon} = \tilde{\Delta}(k^2) - \tilde{\Delta}(k^2) [i\pi] \tilde{\Delta}(k^2) + \dots$$

Def: A one-particle irreducible diagram

is one in which it is still connected after removing any one internal line

at the any one line is cut.

1PI contributions to $i\pi(k^2)$ at $\mathcal{O}(g^4)$



$$\tilde{\Delta}(k^2) = \frac{1}{k^2 + m^2 - i\varepsilon - i\pi(k^2)}$$

* The exact (physical) propagator has a pole at $k^2 = -m^2$ w/ residue one
 • Only consistent if we have

$$\begin{aligned} \pi(-m^2) &= 0 \\ \frac{d}{dk^2} \pi(-m^2) &= \pi'(-m^2) = 0 \end{aligned}$$

→ Use to fix the counter term values for A and B

* We would like to calculate the $i\pi(k^2)$ contribution at $\mathcal{O}(g^2)$

$$\int \frac{d^d l}{(2\pi)^d} \tilde{\Delta}((l+k)^2) \tilde{\Delta}(l^2) \sim \frac{\mathcal{O}(l^4)}{\mathcal{O}(l^4)} \sim \mathcal{O}(l^{d-4})$$

* If $d-4 > 0$ then the integral diverges for large $l \rightarrow \infty$

• Assume $d \leq 4$:

Feynman's formula: $\frac{1}{A_1 \dots A_n} = \int dF_n(x, A_1 + \dots + x_n A_n)^{-n}$

x_i : Feynman parameters

and $\int dF_n = (n-1)! \int_0^1 dx_1 \dots dx_n \delta(x_1 + \dots + x_n - 1)$
 $\hookrightarrow x_1 + \dots + x_n - 1 = 0$
s.t. $\int dF_n 1 = 1$

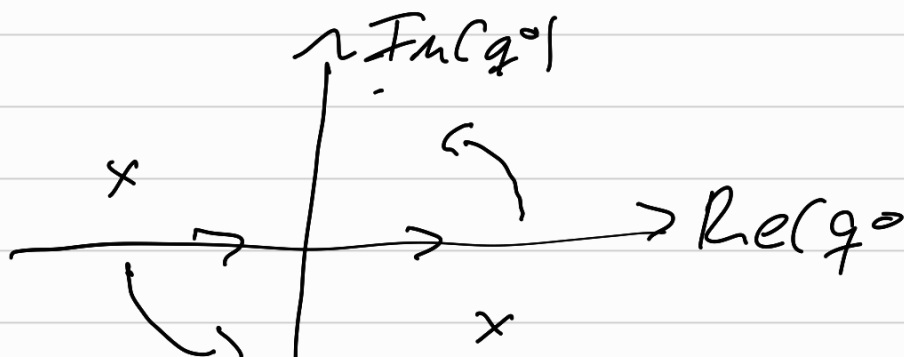
• Here $\tilde{\Delta}((l+k)^2) \tilde{\Delta}(l^2) = \frac{1}{(l^2+m^2)((l+k)^2+m^2)}$
 $\tilde{\Delta} \tilde{\Delta} = \int dF_n (x_2(l^2+m^2) + x_2((l+k)^2+m^2))^{-2}$
but $x_1 + x_2 = 1 \Rightarrow x_2 = 1-x$
 $\tilde{\Delta}((l+k)^2) \tilde{\Delta}(l^2) = \int dF_n (x((l+k)^2+m^2) + (1-x)(l^2+m^2))^{-2}$
 $= \int dF_n [x(l^2 + 2l \cdot k + k^2 + m^2) + l^2 + m^2 - x l^2 - x k^2]$
 $= \int dF_n (x 2l \cdot k + x k^2 + l^2 + m^2)^{-2}$
 $= \int dF_n (x 2l \cdot k + x k^2 + x^2 k^2 - x^2 k^2 + l^2 + m^2)^{-2}$
 $= \int dF_n ((l^2 + x k)^2 + x(1-x)k^2 + m^2)^{-2}$

$\therefore \boxed{q \equiv l + xk}, \quad \boxed{D \equiv x(1-x)k^2 + m^2}$

$\tilde{\Delta}((l+k)^2) \tilde{\Delta}(l^2) = \int_0^1 dF_n (q^2 + D)^{-2}$
• change int. variables

$d^d l = d^d q$

$i\pi(k^2) = \frac{1}{2}(ig^2)\left(\frac{1}{i}\right)^2 \int \frac{d^d q}{(2\pi)^d} \int_0^1 dF_n (q^2 + D)^{-2} + \dots$



the integral over q^0 from $-\infty$ to ∞ is seen as a contour in $\mathbb{C}(q^0)$ plane.
 Assuming $\lim_{|q^0| \rightarrow \infty} i\pi(k^2) = 0$

We can Wick rotate s.t. $q^0 \rightarrow i q^0$ so that the theory is Euclidean

$$\bar{q} \in \mathbb{R}^d \quad \text{s.t.} \quad \bar{q} = (q_1, \dots, q_{d-1}, -i q^0)$$

$$q^0 = i q_d \quad \text{s.t.} \quad \bar{q}^2 = q^2$$

• The measure

$$d^d q = dq^0 dq^1 \dots dq^{d-1} = i d^d \bar{q}$$

$[d^d q = i d^d \bar{q}]$

$$* \int d^d q f(q^2 - i\varepsilon) = i \int d^d \bar{q} f(\bar{q}^2)$$

So long as $f(\bar{q}^2) \rightarrow 0$ faster than $1/\bar{q}^d$ as $\bar{q} \rightarrow \infty$
 Therefore, since $[q^2 + D]^{-2} \sim q^{-4}$ and $\rightarrow 0$ faster than $\bar{q}^d$ ($d < 4$) we have:

$$\pi(k^2) = \frac{1}{2} g^2 I(k^2) - A k^2 - B m^2 + \mathcal{O}(g^4)$$

$$I(k^2) = \int_0^1 dx \int \frac{d^d \bar{q}}{(2\pi)^d} \frac{1}{(q^2 + D)^2}$$

• Can solve this in spherical coordinates

* Differentiate $\pi(k^2)$ twice w.r.t. k^2 to get

$$\Pi''(k^2) = \frac{1}{2} g^2 I''(k^2) + \mathcal{O}(g^4)$$

$$D = \alpha k^2 + m^2 \Rightarrow \frac{d}{dk^2} = \alpha$$

$$I'(k^2) = \int_0^1 dx \int \frac{d^d \bar{q}}{(2\pi)^d} \frac{-2\alpha}{(q^2 + D)^3}$$

$$I''(k^2) = \int_0^1 dx \alpha^2 \int \frac{d^d \bar{q}}{(2\pi)^d} \frac{1}{(q^2 + D)^4}$$

$$\therefore \underline{I''(k^2) = \int_0^1 dx \, 6x^2(1-x)^2 \int \frac{d^d \bar{q}}{(2\pi)^d} \frac{1}{(q^2 + D)^4}}$$

- Now, we can do this integral, integrate twice wrt. k^2 to obtain A and B subject to the B.C.'s

$$\Pi(-m^2) = 0$$

$$\Pi'(-m^2) = 0$$

* The integral is finite for $d < 8$ so this includes $d=6$ where $g\phi^3$ is dimensionless

Taylor expand $\Pi(k^2)$ about $k^2 = -m^2$

$$\begin{aligned} \Pi(k^2) = & \left[\frac{1}{2} g^2 I(-m^2) + (A-B)m^2 \right] \\ & + \left[\frac{1}{2} g^2 I'(-m^2) - A \right] (k^2 + m^2) \\ & + \frac{1}{2!} \left[\frac{1}{2} g^2 I''(-m^2) \right] (k^2 + m^2)^2 + \dots \mathcal{O}(g^4) \end{aligned}$$

- $I(-m^2)$ divergent for $d \geq 4$
- $I'(-m^2)$ divergent for $d \geq 6$
- $I''(-m^2)$ divergent for $d \geq 4 + 2n$

* Use $\mathcal{O}(g^2)$ A and B terms to cancel $\frac{1}{2} g^2 I(-m^2)$ and $\frac{1}{2} g^2 I'(-m^2)$

- Instead of Π'' method we will instead solve $\Pi(k^2)$ directly as a fn of $d < 4$

then take $d \rightarrow d+4$
 (Analytic continuation)
 → This is dimensional regularization

Aside UV-divergences in ϕ^3

$$\langle \Omega | \phi(x) | \Omega \rangle = \frac{\delta}{\delta J} W_1(J) \Big|_{J=0}$$

$$\sim \int \Delta_{xx} \Delta_{yy} = \sum_{\text{tadpoles}} \left(\text{---} \bigcirc \text{---}, \text{---} \bigcirc \text{---}, \text{---} \bigcirc \text{---} \bigcirc \text{---}, \text{---} \bigcirc \text{---} \bigcirc \text{---} \right)$$

$$\langle \Omega | \phi(x) | \Omega \rangle = \frac{1}{2} i g \int d^4 y \left(\frac{1}{i} \Delta_{xy} \Delta_{yy} \frac{1}{i} \right) + \mathcal{O}(g^3)$$

$$\neq 0$$

Thus, we can introduce the counter term
 $\mathcal{L}_\phi$

• If we add $\mathcal{L}_\phi = \mathcal{L}_\phi$ to $\mathcal{L}_{int}$

$$\langle 0 | T \{ \phi(x) \} e^{i \int d^4 w \mathcal{L}_\phi + i \mathcal{L}_{int}} | 0 \rangle$$

$$= \langle 0 | T \{ \phi(x) \} \left(\mathcal{L} + i \int d^4 w \left(\mathcal{L}_\phi(w) + \frac{g}{3!} \phi^3(w) \right) \right)$$

$$= i \int d^4 w \Delta_{xw} \mathcal{L} + i \int d^4 w g \Delta_{xw} \Delta(0)$$

$$\therefore \langle \Omega | T \{ \phi(x) \} | \Omega \rangle = \left(i \mathcal{L} + \frac{1}{2} (i g) \frac{\Delta(0)}{i} \right) \int d^4 y \frac{1}{i} \Delta_{xy} + \mathcal{O}(g^3)$$

• For order to have

$$\langle \Omega | T \{ \phi(x) \} | \Omega \rangle = 0$$

we must therefore have

$$\left(i \mathcal{L} + \frac{1}{2} (i g) \frac{\Delta(0)}{i} + \mathcal{O}(g^3) \right) = 0$$

$$Y = \frac{1}{2} (ig) \Delta(0) + \mathcal{O}(g^3)$$

and so

$$Y = \frac{1}{2} (ig) \Delta(0) + \mathcal{O}(g^3)$$

* Y must be a REAL #; it is the coeff. of a Herm. op. in the Hamiltonian. As such $\Delta(0)$ must be purely imaginary

$$\Delta(0) = \int \frac{d^d k}{(2\pi)^d} \frac{1}{k^2 + m^2 - i\epsilon} \xrightarrow{k \rightarrow \infty} \infty$$

(Ultraviolet-divergence)

* Introduce an UV cutoff Λ :
 $\Lambda \gg m$

$$\Delta(x-y) \rightarrow \int \frac{d^d k}{(2\pi)^d} \frac{e^{ik(x-y)}}{k^2 + m^2 - i\epsilon} \left(\frac{\Lambda^2}{k^2 + \Lambda^2 - i\epsilon} \right)^2$$

We will see shortly that $\Delta(0) = \frac{i}{16\pi^2} \Lambda^2$

* Coming back to $\chi(p^2)$ we must have

$$\chi(p^2) \rightarrow \left(\frac{1}{p^2 + m^2 - i\epsilon} \right) \left(\frac{\Lambda^2}{p^2 + \Lambda^2 - i\epsilon} \right)$$

Here we only add () and not ()² since the other $\chi(p^2)$ will also have () and all be convergent for

$$\frac{1}{p^2 p^2 p^2 p^2} \sim \frac{1}{p^8} \Rightarrow \frac{\int d^d p}{\int p^8} \sim \mathcal{O}(p^{d-8})$$

$$d-8 < 0 \Rightarrow \underline{d < 8}$$

$\frac{1}{\Lambda^2}$ Pauli-Villars

$(p^2 + \Delta^2 - i\epsilon)^{-1}$ Regularization

* Take $\Delta \rightarrow \infty$ at the end.

Thus we have three methods for computing $\Pi(k^2)$

(i) dim. reg.

(ii) Pauli-Villars Reg

(iii) $\Pi''(k^2)$ method

-D Text ops for dim reg.

$$(i) I(k^2) \equiv \int_0^1 dx \int \frac{d^d \vec{q}}{(2\pi)^d} \frac{1}{(\vec{q}^2 + D)^2}$$

Expand into spherical coordinates

$$d^d \vec{q} = (d|\vec{q}|) |\vec{q}|^2 d\Omega_{d-1}$$

$$|\vec{q}| = r$$

$$I(k^2) = \left(\int_0^1 dx \int d|\vec{q}| |\vec{q}|^2 \frac{1}{(|\vec{q}|^2 + D)^2} \right) \underbrace{\left(\int d\Omega_{d-1} \right)}$$

Area of unit sphere in d-dim

$$\Omega_{d-1} = \frac{2\pi^{d/2}}{\Gamma(\frac{d}{2})}$$

(See SUGRA app 22)

$$I(k^2) = \frac{2\pi^{d/2}}{\Gamma(\frac{d}{2})} \int_0^1 dx \int_0^\Lambda d|\vec{q}|$$

$$\Gamma(n+1) = n!$$

$$\Gamma(1) = 1$$

$$\Gamma(-n+x) = \frac{(-1)^n}{n!} \left(\frac{1}{x} - \gamma + \sum_{k=1}^n \frac{1}{k} + O(x) \right)$$

$$\Gamma(z) = \int_0^\infty dt \, t^{z-1} e^{-t}$$

General formula:

$$\int \frac{d^d \vec{q}}{(2\pi)^d} \frac{(\vec{q}^2)^a}{(\vec{q}^2 + D)^b} = \frac{\Gamma(b-a-\frac{1}{2}d) \Gamma(a+\frac{1}{2}d)}{(4\pi)^{d/2} \Gamma(b) \Gamma(\frac{1}{2}d)} \times D^{-(b-a-d/2)}$$

Here $a=1$, $b=2$

* Now, for ϕ^3 : $[\phi] = \frac{1}{2}(d-2)$

$$\therefore -d + \frac{3}{2}(d-2) + [g] = 0$$

$$[g] = d - \frac{3}{2}(d-2) = -\frac{1}{2}d + 3$$

$$[g] = \frac{1}{2}(6-d)$$

thus, ϕ^3 is dimensionless if

$$\epsilon = 6-d = 0$$

• Introduce a new parameter $\tilde{\mu}$ w dim of mass: $[\tilde{\mu}] = +1$

s.t

$$g \rightarrow g \mu^{\epsilon/2}$$

* So that g remains dimensionless $\forall \epsilon$
 $\rightarrow \tilde{\mu}$ is not a free parameter, no physical object (cross-sec.) can depend on it.

* $a=0$, $b=2$, $d=6-\epsilon$ *

$$\frac{\Gamma(2 - 0 - \frac{1}{2}(6 - \epsilon)) \Gamma(\cancel{0} + \frac{1}{2}(6 - \epsilon))}{(4\pi)^{6-\epsilon/2} \Gamma(2) \Gamma(\cancel{\frac{1}{2}(6 - \epsilon)})} D^{-(2 - 0 - \frac{1}{2}(6 - \epsilon))}$$

$$= \frac{\Gamma(-1 + \frac{\epsilon}{2})}{\Gamma(2) (2\pi)^{6-\epsilon/2}} D^{+(-1 + \frac{\epsilon}{2})} \left(\frac{(2\pi)^{\epsilon/2}}{D^{\epsilon/2}} \right) \left\{ \Gamma(2) = 1! \right.$$

$$= \frac{\Gamma(-1 + \frac{\epsilon}{2})}{(4\pi)^3} D \frac{(2\pi)^{\epsilon/2}}{D^{\epsilon/2}}$$

$$\therefore I(k^2) = \frac{\Gamma(-1 + \frac{\epsilon}{2})}{(4\pi)^3} \int_0^1 dx D \left(\frac{4\pi}{D} \right)^{\epsilon/2}$$

Q: Not sure why $a=0$ when the "r" integral $\int d|q| |q|^a / (|q|^2 + D)$

isn't $a=1$ since the Jacobian has $\int d|q| |q|^2$

$$A: \int dK_E \frac{K_E^a}{(K_E^2 + D)^b} = D^{\frac{d+1}{2}} \frac{\Gamma[\frac{a+1}{2}] \Gamma(b - \frac{a+1}{2})}{2 \Gamma(b)}$$

$$b = a = 2$$

$$I(k^2) = \frac{\Gamma(-1 + \frac{\epsilon}{2})}{(4\pi)^3} \int_0^1 dx D \left(\frac{4\pi}{D} \right)^{\epsilon/2}$$

$$g \rightarrow g \tilde{\mu}^{\epsilon/2} \quad | \quad \alpha \equiv g^2 / (4\pi)^3$$

$$\Pi(k^2) = \frac{1}{2} g^2 I(k^2) - A k^2 - B m^2 + \mathcal{O}(g^4)$$

$$\Pi(k^2) = \frac{1}{2} g^2 \tilde{\mu}^{\epsilon} \left[\frac{\Gamma(-1 + \epsilon/2)}{(4\pi)^3} \int_0^1 dx D \left(\frac{4\pi}{D} \right)^{\epsilon/2} \right] - A k^2 - B m^2 + \mathcal{O}(g^4)$$

$$\Pi(k^2) = \frac{1}{2} \alpha \Gamma(-1 + \epsilon/2) \int_0^1 dx D \left(\frac{4\pi \tilde{\mu}^2}{D} \right)^{\epsilon/2}$$

* Now take the limit $\epsilon \rightarrow 0$:

(i)

$$f(\epsilon) = M^{\epsilon/2}$$

$$f'(\epsilon) = \frac{d}{d\epsilon} (e^{\frac{\epsilon}{2} \ln(M)}) = \frac{1}{2} M^{\epsilon/2} \ln(M)$$

$$f'(0) = \frac{1}{2} \ln(M)$$

$$\therefore f(\epsilon) = f(0) + f'(0)\epsilon + \frac{1}{2} f''(0)\epsilon^2$$

$$M^{\epsilon/2} = 1 + \frac{\epsilon}{2} \ln(M) + \mathcal{O}(\epsilon^2)$$

which we will use for $()^{\epsilon/2}$

$$\int_0^1 dx D \left[1 + \frac{\epsilon}{2} \ln\left(\frac{4\pi\tilde{\mu}^2}{D}\right) + \mathcal{O}(\epsilon^2) \right]$$

$$= \int_0^1 dx D + \frac{\epsilon}{2} \int_0^1 dx D \ln\left(\frac{4\pi\tilde{\mu}^2}{D}\right) + \mathcal{O}(\epsilon)$$

$$D = (x - x^2) k^2 + m^2$$

$$\rightarrow \int_0^1 dx D = \left(\frac{1}{2} - \frac{1}{3}\right) k^2 + m^2 = \frac{1}{6} k^2 + m^2$$

$$\int_0^1 dx D \left(\frac{4\pi\tilde{\mu}^2}{D}\right)^{\epsilon/2} = \frac{1}{6} k^2 + m^2 + \frac{\epsilon}{2} \int_0^1 dx D \ln\left(\frac{4\pi\tilde{\mu}^2}{D}\right)$$

(ii)

$$\Gamma(-1 + \frac{\epsilon}{2}) = (-1) \left[\frac{2}{\epsilon} - \gamma + 1 + \mathcal{O}(\frac{\epsilon}{2}) \right]$$

Putting (i) and (ii) together we find

$$\frac{1}{2} \alpha \left[(-1) \left\{ \frac{2}{\epsilon} - \gamma + 1 + \mathcal{O}(\epsilon/2) \right\} \right] \left(\left(\frac{1}{6} k^2 + m^2 \right) + \frac{\epsilon}{2} \int_0^1 dx \right)$$

(* All terms $\mathcal{O}(\epsilon)$ and greater will vanish in the limit $\epsilon \rightarrow 0$)

$$= \left(-\frac{\alpha}{2}\right) \left[\int_0^1 dx D \ln \left(\frac{4\pi \tilde{\mu}^2}{D} \right) + \left(\frac{2}{\epsilon} + 1\right) \left(\frac{1}{6} k^2 + m^2 \right) - \alpha \int_0^1 dx D \right]$$

↪ this is $(\frac{1}{6} k^2 + m^2)$

$$= \ln(e^{-\gamma}) \int_0^1 dx D$$

$$= \int_0^1 dx D \ln(e^{-\gamma})$$

This along w yields

$$\int_0^1 dx D \left[\ln \left(\frac{4\pi \tilde{\mu}^2}{D} \right) + \ln(e^{-\gamma}) \right]$$

$$= \int_0^1 dx D \ln \left(\frac{4\pi \tilde{\mu}^2}{D e^{\gamma}} \right)$$

Putting this all together we find:

$$\Pi(k^2) = \left(-\frac{\alpha}{2}\right) \left[\left(\frac{2}{\epsilon} + 1\right) \left(\frac{1}{6} k^2 + m^2 \right) + \int_0^1 dx D \ln \left(\frac{4\pi \tilde{\mu}^2}{D e^{\gamma}} \right) \right]$$

$$= -A k^2 - B m^2 + \mathcal{O}(\alpha^2)$$

Define: $\mu \equiv \sqrt{4\pi} e^{-\gamma/2} \tilde{\mu}$

s.t. $\mu^2 = 4\pi \tilde{\mu}^2 e^{-\gamma}$

$$\ln \left(\frac{4\pi \tilde{\mu}^2}{D e^{\gamma}} \right) = \ln(\mu^2 / D)$$

$$-\frac{\alpha}{2} \int_0^1 dx D \left[\ln(\mu^2) - \ln(D) \right]$$

$$= -\frac{\alpha}{2} \ln(\mu^2) \left(\frac{1}{6} k^2 + m^2 \right) + \frac{\alpha}{2} \int_0^1 dx D \ln(D)$$

$$= -\frac{\alpha}{6} \ln(\mu) k^2 - \alpha \ln(\mu) m^2 + \frac{\alpha}{2} \int_0^1 dx D \ln(D/m^2)$$

$$+ \alpha \left(\frac{1}{6} k^2 + m^2 \right) \ln(\mu)$$

$$= k^2 \frac{\alpha}{6} \left[-\ln(\mu) + \ln(\mu) \right] + m^2 \left(-\alpha \ln(\mu) + \alpha \ln(\mu) \right)$$

$$+ \alpha/2 \int_0^1 dx D \ln(D/m^2)$$

$$\therefore \frac{\alpha}{2} \int_0^1 dx D \ln(D) = \alpha \left(\frac{1}{\epsilon} + 1 \right) \left(\frac{1}{6} k^2 + m^2 \right)$$

$$\Pi(k^2) = \frac{\alpha}{2} \int_0^1 dx D \ln\left(\frac{D}{m^2}\right) - K^2 \left(\frac{\alpha}{6}\right) [\ln(\mu/m)] - m^2 \alpha \ln(M/m)$$

$$\Pi(k^2) = \frac{\alpha}{2} \int_0^1 dx D \ln\left(\frac{D}{m^2}\right) - K^2 \left\{ \frac{\alpha}{6} \left[\frac{1}{2} + \ln(\mu/m) + \frac{1}{2} \right] + A \right\} - m^2 \left\{ \alpha \left[\frac{1}{2} + \ln(\mu/m) + \frac{1}{2} \right] + B \right\} + \mathcal{O}(\alpha^2)$$

* Choose A and B s.t.

$$A = -\frac{1}{6} \alpha \left[\frac{1}{2} + \ln\left(\frac{\mu}{m}\right) + \frac{1}{2} + K_A \right] + \mathcal{O}(\alpha^2)$$

$$B = -\alpha \left[\frac{1}{2} + \ln(\mu/m) + \frac{1}{2} + K_B \right] + \mathcal{O}(\alpha^2)$$

$$K_A, K_B \in \mathbb{R}$$

$$\Pi(k^2) = \frac{\alpha}{2} \int_0^1 dx D \ln\left(\frac{D}{m^2}\right) + \alpha \left(\frac{1}{6} K_A k^2 + K_B m^2 \right) + \mathcal{O}(\alpha^2)$$

• Find K_A and K_B via B.C.'s: $\Pi(-m^2) = 0$
 $\Pi'(-m^2) = 0$

$$D_0 = D|_{k^2 = -m^2} = x(1-x)(-m^2) + m^2 = m^2 [1 - x(1-x)]$$

$$\frac{\alpha}{2} \int_0^1 dx D_0 \ln\left(\frac{D_0}{m^2}\right) + \alpha m^2 \left(K_B - \frac{1}{6} K_A \right) = 0$$

$$\frac{\alpha}{2} \int_0^1 dx D_0 \ln(D_0) = \alpha \int_0^1 dx D_0 \ln(m) + \alpha$$

$$\left(\int_0^1 dx (1 - x + x^2) = 1 - \frac{1}{2} + \frac{1}{3} = \frac{5}{6} \right.$$

$$\left. \int_0^1 dx [1 - x(1-x)] \ln(1 - x(1-x)) \right.$$

$$\int_0^1 dx [1 - x(1-x)] \ln(1 - x(1-x))$$

$$= \left[x - \frac{1}{2}x^2 + \frac{1}{3}x^3 \right] \ln[1 - x(1-x)] \Big|_0^1 - \int dx \left(\frac{x - \frac{1}{2}x^2 + \frac{1}{3}x^3}{1-x+x^2} \right) (2x-1)$$

Can impose $\overline{U}(-m^2) = 0$ via

$$\Pi(k^2) = \frac{1}{2}\alpha \int_0^1 dx D \ln\left(\frac{D}{D_0}\right) + \text{linear in } (k^2 + m^2)$$

$$\therefore \frac{1}{2}\alpha \int_0^1 dx \left[x(1-x) \right] \ln\left(\frac{D}{D_0}\right) \Big|_{k^2=m^2} + \frac{1}{2}\alpha \int_0^1 dx D \frac{D_0}{D} \left[x(1-x) \right] \frac{1}{D_0} + W = 0$$

$$\frac{\alpha}{2} \int_0^1 dx [x(1-x)] + W = 0$$

$$W = -\frac{\alpha}{2} \left[\frac{1}{2}x^2 - \frac{1}{3}x^3 \right]_0^1 = -\frac{\alpha}{12}$$

$$\therefore W = -\alpha/12$$

$$\Pi(k^2) = \frac{1}{2}\alpha \int_0^1 dx D \ln\left(\frac{D}{D_0}\right) - \frac{\alpha}{12}(k^2 + m^2) + \mathcal{O}(\alpha^2)$$

$$* \int_0^1 dx [k^2 x(1-x) + m^2] \ln\left(\frac{k^2 x(1-x) + m^2}{m^2(1-x(1-x))}\right)$$

$$\int_0^1 dx [k^2 x(1-x) \ln(D)] - \int_0^1 dx k^2 x(1-x) \ln(D_0) + \int_0^1 dx m^2 \ln(D) - \int_0^1 dx m^2 \ln(D_0)$$

$$\textcircled{a} \int_0^1 k^2 x(1-x) \ln(k^2 x(1-x) + m^2) dx$$

... just use Wolfram ...

$$\Pi(k^2) = \frac{1}{2}\alpha \left[(3 - \pi\sqrt{3})k^2 + (3 - 2\pi\sqrt{3})m^2 + 2k^2 \left(1 + \frac{4m^2}{k^2} \right)^{3/2} + \tanh^{-1} \left(\frac{1}{\sqrt{1 + 4m^2/k^2}} \right) \right] + \mathcal{O}(\alpha^2)$$

• Square root branch pt at $k^2 = -4m^2$ at which point $\pi(k^2)$ assumes an imaginary part for $k^2 < -4m^2$

Now, the full interacting propagator is

$$\hat{\Delta}(k^2) = \frac{1}{k^2 + m^2 - i\epsilon - \pi(k^2)} = \left(\frac{1}{1 - \pi(k^2)/(k^2 + m^2)} \right) \left(\frac{1}{k^2 + m^2 - i\epsilon} \right)$$

$$\tanh^{-1}(z) = \frac{1}{2} [\ln(1+z) - \ln(1-z)]$$

14.1 $\Gamma(\alpha) = \int_0^\infty dt t^{\alpha-1} e^{-t}$
 $\Gamma(\alpha)/A^\alpha = \int_0^\infty dt A^{-\alpha} t^{\alpha-1} e^{-t}$

$$\begin{aligned} & t \rightarrow tA \\ & = \int_0^\infty A dt A^{-\alpha} (tA)^{\alpha-1} e^{-tA} \\ & = \int_0^\infty dt A^{1-\alpha} A^{\alpha-1} t^{\alpha-1} e^{-tA} \\ & = \int_0^\infty dt t^{\alpha-1} e^{-tA} \end{aligned}$$

as claimed

$$\frac{\Gamma(\alpha_i)}{A_i^{\alpha_i}} = \int_0^\infty dt_i t_i^{\alpha_i-1} e^{-A_i t_i}$$

$$\text{Thus } \prod_{i=1}^n \left(\frac{\Gamma(\alpha_i)}{A_i^{\alpha_i}} \right) = \prod_{i=1}^n \left(\int_0^\infty dt_i t_i^{\alpha_i-1} e^{-A_i t_i} \right)$$

$$\frac{\prod_{i=1}^n \Gamma(\alpha_i)}{\prod_{i=1}^n A_i^{\alpha_i}} = \prod_{i=1}^n \left[\int_0^\infty dt_i t_i^{\alpha_i-1} e^{-A_i t_i} \int_0^\infty ds \delta(s - \sum_{i=1}^n t_i) \right]$$

$$t_i = s x_i$$

$$\frac{\prod_{i=1}^n \Gamma(\alpha_i)}{\prod_{i=1}^n A_i^{\alpha_i}} = \int_0^\infty ds \delta(s(1 - \sum_{i=1}^n x_i)) s^n \prod_{i=1}^n \int_0^\infty dx_i (s x_i)^{\alpha_i-1} e^{-A_i s x_i}$$

$$\begin{aligned}
 \text{Since } \delta(SX) &= \frac{\delta(x)}{|S|} \\
 &= \int_0^\infty ds \delta(1 - \sum_{i=1}^n x_i) s^{n-1} \left(\prod_{i=1}^n \int_0^\infty dx_i \right) s^{\sum(\alpha_i - 1)} x_i^{\alpha_i - 1 - 1} e^{-\sum A_i x_i} \\
 &= \int dx_1 \dots dx_n \delta(1 - \sum_{i=1}^n x_i) \left(\prod_{i=1}^n x_i^{\alpha_i - 1} \right) \int_0^\infty ds s^{n-1} s^{\sum \alpha_i - n} e^{-\sum A_i s x_i} \\
 &= \frac{1}{(n-1)!} \int dF_n \left(\prod_{i=1}^n x_i^{\alpha_i - 1} \right) \int_0^\infty ds s^{\sum \alpha_i - 1} e^{-\sum A_i s x_i} \\
 &\quad \underbrace{\hspace{10em}} \\
 &\quad \frac{\Gamma(\sum \alpha_i)}{\left(\sum_{i=1}^n A_i x_i \right)^{\sum \alpha_i}}
 \end{aligned}$$

Thus, the general form of Feynman's formula is given by:

$$\frac{1}{A_1^{\alpha_1} \dots A_n^{\alpha_n}} = \frac{\Gamma(\sum \alpha_i)}{\prod \Gamma(\alpha_i)} \frac{1}{(n-1)!} \int dF_n \frac{\prod_{i=1}^n (x_i^{\alpha_i - 1})}{\left(\sum_{i=1}^n x_i A_i \right)^{\sum \alpha_i}}$$

14.3 $\int d^d q q^\mu f(q^2)$

By Lorentz invariance the left side can also only be a vector

$$\int d^d q q^\mu f(q^2) = A V^\mu$$

However if we take $q^\mu \rightarrow -q^\mu$

$$\int d^d q (-q^\mu) f(q^2) = -A V^\mu$$

$$\text{Thus, } A V^\mu = -A V^\mu \Rightarrow A = 0$$

i.e. this integral is odd.

(ii) likewise

$$\int d^d q q^\mu q^\nu f(q^2) = A g^{\mu\nu} + B k^\mu V^\nu$$

However it cannot depend on any 4-vector

$$\text{so } B = 0$$

$$g_{\mu\nu} \int d^d q g^\mu q^\nu f(q^2) = A d$$

$$\therefore A = \frac{1}{d} \int d^d q q^2 f(q^2)$$

$$\int d^d q g^\mu q^\nu f(q^2) = (g^{\mu\nu}/d) \int d^d q q^2 f(q^2)$$

$$\textcircled{b} \int d^d q g^\mu q^\nu q^\rho q^\sigma f(q) = g^{\mu\nu} g^{\rho\sigma} A + g^{\mu\rho} g^{\nu\sigma} B + g^{\mu\sigma} g^{\nu\rho} C$$

Since it must be symmetric among all indices.

Contract with each A, B, C coeff.

$$A = B = C = \frac{1}{2d(d+2)} \int d^d q q^4 f(q)$$

Compute $\mathcal{O}(x)$ corrections to the prop in ϕ^4 theory

$$\mathcal{L}_2 = -\frac{1}{24} Z_\lambda \lambda \phi^4$$

$$\langle \Omega | T \phi_1 \phi_2 | \Omega \rangle_c = \langle 0 | T \phi_1 \phi_2 e^{i \int d^d w \mathcal{L}_2} | 0 \rangle$$

$$= Z_0 | \phi_1 \phi_2 (-i \frac{1}{24} Z_\lambda) \lambda \int d^d w \phi_w^4 | 0 \rangle$$

$$\int \frac{d^d \ell}{(2\pi)^4} (-i\lambda) \left(\frac{(-i)}{\ell^2 + m^2 - i\epsilon} \right) \left(\frac{1}{2} \right) \leftarrow \text{Sym factor}$$

$$= \frac{1}{(2\pi)^4} \int d^d \ell \frac{1}{\ell^2 + m^2 - i\epsilon} = -i/(4\pi^2 + B m^2)$$

$$\therefore i\Gamma = ((-i)^{d/2} / 2) \lambda \int (2\pi)^d \ell^{d/2} \ell^2 \ell^{d/2-1} d\ell$$

$$+ \mathcal{O}(\lambda^2)$$

$$\Gamma = \frac{i\lambda}{2} \int \frac{d^d \ell}{(2\pi)^d} \left(\frac{1}{\ell^2 + m^2 - i\epsilon} \right) - (A\ell^2 + Bm^2) + \mathcal{O}(\lambda^2)$$

Note this integral is $[J]^{d-2}$
and diverges if $d-2 > 0$

Wick rotate $q_0 \rightarrow i\bar{q}_0$

$$\int d^d q f(q^2 - i\epsilon) = i \int d^d \bar{q} f(\bar{q}^2)$$

$$-\frac{\lambda}{2} \int \frac{d^d \bar{\ell}}{(2\pi)^d} \frac{1}{(\bar{\ell}^2 + m^2)}$$

Now in ϕ^4 : $[\phi] = \frac{1}{2}(d-2)$

$$[\phi^4] = 2(d-2)$$

$$[d^d x] = d$$

$$[\lambda] = d + 2d - 4 = 0$$

$$[\lambda] = 4 - d$$

$$\boxed{d = 4 - d}$$

$\lambda \rightarrow \lambda \tilde{\mu}^{\epsilon}$ ϵ : dimensionless

$$[\tilde{\mu}^{\epsilon}] = 4 - d$$

$$-\frac{\lambda \tilde{\mu}^{\epsilon}}{2} \int \frac{d^d \bar{\ell}}{(2\pi)^d} \frac{1}{\bar{\ell}^2 + m^2}$$

$$\lambda \tilde{\mu}^{\epsilon} / \Gamma(1 - \frac{d}{2}) \Gamma(\frac{d}{2}) \sim (1 - d/2)$$

$$= \frac{-\lambda \tilde{\mu}^2}{2} \left(\frac{\Gamma(1 - \frac{\epsilon}{2}) \Gamma(\frac{1}{2}d)}{(4\pi)^{d/2} \Gamma(\frac{1}{2}) \Gamma(\frac{1}{2}d)} \right) (M^2)^{\frac{(4-\epsilon)/2}{2} - 1} = -i$$

$$= -\frac{\lambda \tilde{\mu}^2}{2} \left(\frac{\Gamma(1 - \frac{1}{2}(4-\epsilon))}{(4\pi)^{4-\epsilon/2}} \right) M^2^{\frac{(4-\epsilon)/2}{2} - 1}$$

$$= -\frac{\lambda \tilde{\mu}^2}{2} \frac{\Gamma(\frac{\epsilon}{2} - 1) (4\pi)^{\epsilon/2}}{(4\pi)^2} \frac{(M^2)}{(M^2)^{\epsilon/2}}$$

$$= \left(\frac{-\lambda \tilde{\mu}^2 M^2}{(4\pi)^2 2} \right) \Gamma\left(\frac{\epsilon}{2} - 1\right) \left(\frac{4\pi}{M^2}\right)^{\epsilon/2}$$

Now, $A^\epsilon = 1 + \frac{\epsilon}{2} \ln(A) + \mathcal{O}(\epsilon^2)$

$$\Gamma\left(\frac{\epsilon}{2} - 1\right) = (-1) \left[\frac{2}{\epsilon} - \gamma + 1 + \mathcal{O}\left(\frac{\epsilon}{2}\right) \right]$$

$$A = \frac{4\pi \tilde{\mu}^2}{m^2}$$

$$= \left(\frac{\lambda m^2}{(4\pi)^2 2} \right) \left[\frac{2}{\epsilon} - \gamma + 1 + \mathcal{O}(\epsilon) \right] \left(1 + \frac{\epsilon}{2} \ln\left(\frac{4\pi \tilde{\mu}^2}{m^2}\right) + \mathcal{O}(\epsilon^2) \right)$$

$$= \left(\frac{\lambda m^2}{(4\pi)^2 2} \right) \left(\frac{2}{\epsilon} - \gamma + 1 + \ln\left(\frac{4\pi \tilde{\mu}^2}{m^2}\right) \right)$$

$$\Pi(k^2) = \left(\frac{\lambda m^2}{(4\pi)^2 2} \right) \left[\frac{2}{\epsilon} - \gamma + 1 + \ln\left(\frac{4\pi \tilde{\mu}^2}{m^2}\right) \right]$$

$$= A k^2 - B m^2 + \mathcal{O}(\lambda^2)$$

Now impose $\Pi(-m^2) = 0$

$$\omega m^2 + (A - B) m^2 = 0$$

$$B = \omega/m^2 + A$$

* Goal is to now take A and $B \propto \lambda$

$$A = \lambda K_A + \mathcal{O}(\lambda^2)$$

$$m^2 \left[\frac{\lambda}{(uv)^2} \left[\frac{2}{\varepsilon} - \gamma + 1 + \ln \left(\frac{4\pi \hat{s}^2}{m^2} \right) \right] - B \right] \\ = \lambda R_B$$

$$T(u^2) = -\lambda K_A u^2 - \lambda K_B m^2 + \mathcal{O}(\lambda^2)$$

$$\Pi(-m^2) = 0 \quad \checkmark$$

$$\pi'(u^2)|_{u^2=0} = 0 = -\lambda u_4$$

and so $u_4 = 0$

$$\Pi(k^2) = -k_B \lambda m^2 + \mathcal{O}(x^2)$$

Reimpose $\Pi(-m^2) \neq 0$
and so $k_B = 0$

$$\overline{\Pi}(x^2) = \mathcal{Q}(x^2)$$

$$A = 0$$

$$B = \frac{m^2 \lambda}{(4\pi)^2 z} \left[\frac{2}{\varepsilon} - \gamma + 1 + \ln\left(\frac{4\pi \tilde{\mu}^2}{m^2}\right) \right] \mathcal{H}[x^2]$$

$$\mu = \sqrt{4\pi} e^{-\gamma/2} \mu$$

$$B = \frac{m}{(4\pi)^2} \left[\frac{2}{\epsilon} + 1 + \ln \left(\frac{\mu^2}{m^2} \right) \right]$$

• For a $\mathbb{C}$ -scalar theory
 $\sim -\frac{1}{4} Z_1 \lambda (\phi^\dagger \phi)^2$ the only difference
 is the symmetry factor $\frac{\pm 1}{2} \rightarrow \frac{1}{2}$

Renormalization of the anharmonic oscillator

$$h = \frac{1}{2} Z \dot{q}^2 - \frac{1}{2} Z \omega^2 q^2 - Z_1 \lambda \omega^3 q^4$$